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Welcome to CANOPIE-HPC!

The 7th *International Workshop on Containers and New Orchestration Paradigms for Isolated Environments in HPC*

Steering Committee:
Andrew J. Younge (Sandia), R. Shane Canon (LBNL/NERSC)

Program Chairs:
Alberto Madonna (CSCS), Claudia Misale (CoreWeave)

https://canopie-hpc.org

Welcome to CANOPIE-HPC

7th International Workshop on **C**ontainers **a**nd **N**ew **O**rchestration **P**aradigms for **I**solated **E**nvironments in **HPC**

- Principal venue for research in cutting-edge container technologies, virtualization, and OS system software as it relates to supporting High Performance Computing (HPC)

- This is our seventh iteration of the workshop
  - Premier full-day SC25 workshop
  - All proceedings to be published by ACM

- Thank you for participating!

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This workshop is possible only with a great Technical Program Committee!

The committee received **12** paper and **7** lightning talk submissions and accepted **7** papers for final publications in the proceedings and **5** lightning talks for presentation

- Carlos Eduardo Arango Gutierrez, NVIDIA Corporation
- Abdulrahman Azab, University of Oslo; PRACE
- Carlos Jaime Barrios Hernandez, SCALAC, Industrial University of Santander (Colombia)
- Angel Beltre, Sandia National Laboratories
- Patrick Bridges, University of New Mexico
- Taylor Childers, Argonne National Laboratory
- Marcin Copik, ETH Zurich
- Francois Diakhate, CEA-DAM
- Holger Gantikow, Atos, Plymouth University
- Joshua Hursey, IBM
- Christian Kniep, QNIB Solutions
- Barbara Krasovec, Jožef Stefan Institute
- Adam Lavely, Lawrence Berkeley National Laboratory
- Zhuoran Liu, IBM T.J. Watson Research Center
- Nina Mujkanovic, Swiss National Supercomputing Centre
- CJ Newburn, NVIDIA Corporation
- Henrik Nortamo, CSC
- Megan Phinney, Los Alamos National Laboratory
- Sameer Shende, University of Oregon
- Mengmei Ye, IBM TJ Watson Research Center

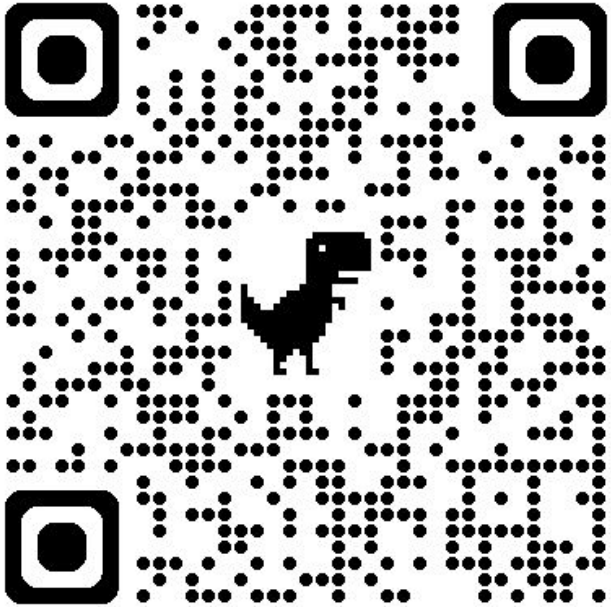
CANOPIE-HPC Morning

Start	Duration	Title	Speaker	Type	Files
09:00 AM		Start			
09:00 AM	15 min	CANOPIE-HPC - Introduction and welcome	Madonna, Misale, Canon, Younge		
09:15 AM	45 min	The Convergence of Containers and Kubernetes with HPC and Generative AI	Gupta, Jackson. Red Hat	Invited Talk	
10:00 AM	30 min	Morning Break			
10:30 AM	30 min	Experience Deploying Containerized GenAI Services at an HPC Center	Beltre, Ogden, Pedretti. Sandia National Laboratories	Paper	
11:00 AM	30 min	Engine-Agnostic Model Hot-Swapping for Cost-Effective LLM Inference	Stoyanov (University of Oxford, Red Hat), Spišaková, Reber, Armour, Copík, Bruno	Paper	
11:30 AM	10 min	To Virtualize or Not to Virtualize: Experiences from Building Two Generations of Virtualized Infrastructure for LLM Training	Mohan, Chen, Chung, Welp, Seelam. IBM Research	Lightning talk	
11:40 AM	10 min	Dynamic Topology-Aware Scheduling in HPC Systems with Topograph	Shmulevich. NVIDIA Corporation	Lightning talk	
11:50 AM	10 min	Evolving Sarus to augment Podman for HPC	Madonna, Cruz, Lee, Chesi. ETH Zurich, CSCS	Lightning talk	
12:00 PM	30 min	Seamless End-to-End Containerized HPC Environments	Cook, Canon, Lavely, Margala. NERSC, Lawrence Berkeley National Laboratory	Paper	
12:30 PM	90 min	Lunch Break			

CANOPIE-HPC Afternoon

2:00 PM	30 min	Usability Evaluation of Cloud for HPC Applications	Sochat, Milroy, Sarkar, Marathe, Patki. Lawrence Livermore National Laboratory	Paper
2:30 PM	30 min	An Elastic Job Scheduler for HPC Applications on the Cloud	Bhosale, Chandrasekar, Kale, Kokkila-Schumacher. University of Illinois at Urbana-Champaign, IBM Research	Paper
3:00	30 min	Afternoon Break		
3:30 PM	10 min	Charliecloud with Nvidia GPUS via standards new and old	Phinney, Loshak. Los Alamos National Laboratory	Lightning talk
3:40 PM	10 min	Charliecloud as a Kubernetes Container Runtime	Loshak, Bielicke. Los Alamos National Laboratory	Lightning talk
3:50 PM	30 min	Towards Enabling Hostile Multi-tenancy in Kubernetes	Kanso, Oks, Mostafa, Virdi. Microsoft Corporation	Paper
4:20 PM	30 min	Evaluating HPK for Running Cloud-Native Workloads on Slurm Clusters	Chazapis, Vassilakis, Petsis, Marazakis, Bilas. Foundation for Research and Technology - Hellas	Paper
4:50 PM	40 min	Community Discussion and Open Q&A	Madonna, Misale, Canon, Younge	Panel
5:30 PM		End		

SC25 Workshops proceedings are online now!



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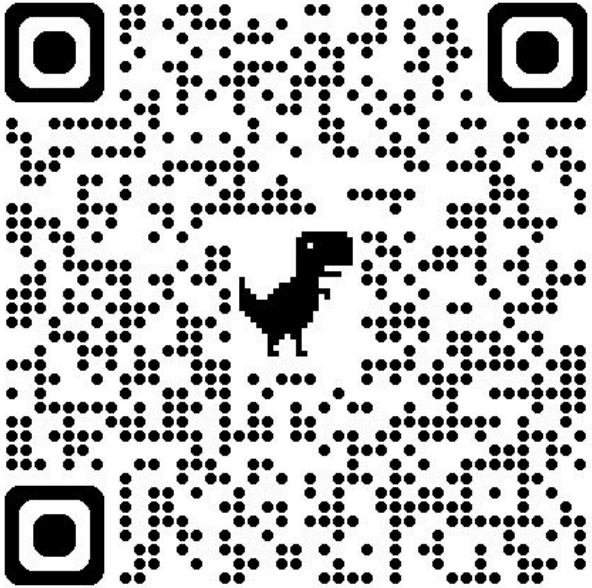
Join the HPC-Containers Slack channel!

Workshop survey & feedback!



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Workshop survey & feedback!